

Directional Disagreement Between Prior Returns and Opening Gaps as a Predictor of Short-Term Equity Option Returns

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Abstract

We examine whether the interaction between prior-day equity returns and same-day opening gaps predicts short-term equity option returns. Contrary to conventional momentum intuition, we find that when prior returns and opening gaps move in opposite directions (*directional disagreement*), subsequent option returns are substantially higher than when both signals align.

Using a simulated dataset of 19,601 at-the-money call option trades across 50 large-cap U.S. equities from 2020 to 2026, disagreement conditions produce an average return of approximately +32% per trade with an 82% win rate, compared to +7.6% and 51% under aligned conditions. These results persist in an out-of-sample period (2024–2026), where disagreement conditions maintain an 81% win rate and a mean return of +31.8%.

The effect is strongest at the market open and decays monotonically with delayed entry. Feature importance analysis indicates that the interaction between prior returns and opening gaps accounts for the majority of predictive signal across candidate variables.

Results are based on simulated execution with simplified option pricing assumptions and should be interpreted as exploratory. Ongoing live forward testing is expected to provide further validation.

Keywords: options returns, intraday trading, opening gaps, momentum reversal, market microstructure, equity options, short-term trading

Disclosure and Limitations

This study is conducted by an independent researcher and is not affiliated with any financial institution. All results are based on simulated execution and do not reflect actual trading performance.

The simulation relies on simplified option pricing assumptions, including the use of historical volatility as a proxy for implied volatility and fixed transaction cost estimates. Real-world execution may differ materially due to bid-ask spreads, slippage, liquidity constraints, and intraday volatility dynamics.

The dataset is subject to survivorship bias, as it includes only large-cap equities with continuous data availability. These findings are intended for research purposes only and do not constitute investment advice.

1. Introduction

A large body of empirical literature documents that equity momentum — the tendency for recent winners to continue outperforming — is one of the most robust cross-sectional return predictors identified in asset pricing research (Jegadeesh and Titman, 1993). Within the options market, practitioners and researchers alike have generally assumed that directional signals compound: if a stock closed higher yesterday and gaps higher at the open, the case for a bullish options position appears strongest. This paper tests whether that intuition holds at short horizons.

We define a *Directional Disagreement Signal* (DDS) as the condition in which the prior trading day's return is positive while the same-day opening gap — defined as the percentage change between the prior close and the current market open — is flat or negative. This configuration, which we label a *mixed* signal, stands in contrast to the *aligned* condition in which both signals point in the same direction.

Using a simulation-based methodology applied to 50 large-cap U.S. equities over the period 2020–2026, we find that at-the-money call options purchased at the 9:31 AM market open under directional disagreement conditions produce substantially higher average returns and win rates than those purchased under aligned conditions. The finding is robust to out-of-sample testing, permutation analysis, and disaggregation across distinct market regimes.

Importantly, we frame these results as hypothesis-generating rather than conclusive, pending validation under real-world execution conditions. The primary contribution of this paper is to identify a parsimonious, interpretable two-variable interaction that challenges conventional momentum-continuation intuition in the options market and warrants further investigation using live trading data.

2. Related Literature

The predictability of short-term equity returns around the market open has received considerable attention. Gao, Han, Li, and Zhou (2018) document an intraday momentum effect in which the first half-hour return predicts the last half-hour return in U.S. equity markets. Their findings suggest that opening price dynamics contain information relevant to intraday price discovery beyond what is reflected in prior-day returns alone.

Within the options market, Goyal and Saretto (2009) identify that the spread between historical and implied volatility is a significant predictor of option returns in the cross-section. Their work underscores the importance of volatility mispricing as a source of option return variation — a mechanism that may partly underlie the effect documented here, insofar as directional disagreement conditions correspond to episodes of relatively lower implied volatility for at-the-money calls.

The role of opening gaps in equity price dynamics has been examined in the context of gap continuation versus gap reversal. Empirically, overnight gaps in large-cap equities tend to partially revert intraday, particularly when the gap occurs following a strong prior-day directional move. This partial reversion creates conditions in which options markets may misprice near-term directional exposure at the

open. Muravyev and Pearson (2020) provide evidence that effective options trading costs are lower than commonly assumed, lending further credence to simulation-based approaches that account for realistic spread and slippage assumptions.

The momentum reversal literature (Jegadeesh, 1990; Lehmann, 1990) documents that short-horizon return reversals are a pervasive feature of equity markets, attributable to market microstructure effects and inventory management by market makers. The DDS signal identified in this paper may be viewed as an options-market manifestation of short-horizon reversal dynamics, where the opening pullback following a prior-day gain creates a temporary mispricing in near-term call option premiums.

3. Data and Methodology

3.1 Universe and Time Period

The analysis covers 50 large-cap U.S. equities drawn from the S&P 500, selected for continuous data availability over the full sample period. Intraday price data at the one-minute frequency is sourced from a commercial data provider (Polygon.io) for the period January 2020 through April 2026, yielding approximately 1,382 trading days with bar-level coverage. Prior-day return and opening gap variables are computed at the daily frequency using official prior close and same-day market open prices.

The universe is subject to survivorship bias in that all 50 equities maintained continuous large-cap status and data availability across the sample period. Results should be interpreted in light of this constraint, and generalization to smaller-cap or less liquid equities is not implied.

3.2 Option Pricing and Execution Assumptions

Option prices are computed using the Black-Scholes model (Black and Scholes, 1973). For each trade, the strike price is set to the nearest available strike at or above the 9:31 AM market open price, and the expiration is set to seven calendar days forward, consistent with weekly option contracts. Historical 30-day realized volatility is used as a proxy for implied volatility.

A reality layer is applied to all simulated trades to account for execution friction: a 7% bid-ask spread is applied to the theoretical option price at entry, and a 2% slippage factor is applied at exit. These figures are conservative relative to observed spreads for liquid large-cap weekly options but are used to ensure that reported results do not depend on assumptions of frictionless execution.

Trade exits are triggered by one of three conditions: (1) the option price declines by the stop-loss threshold, (2) the option price reaches the profit target, or (3) the trading day concludes without either trigger being reached, at which point the position is closed at the prevailing market price.

3.3 Signal Construction and Limitations

The prior-day return (Ret_{1d}) is defined as the percentage change between the close price on day $t-1$ and the close price on day $t-2$. The opening gap ($OpeningGap$) is defined as the percentage change between the 9:31 AM open price on day t and the close price on day $t-1$. The DDS condition is satisfied when Ret_{1d} is strictly positive and $OpeningGap$ is zero or negative.

A key limitation of this methodology is the use of historical volatility as a substitute for implied volatility. In practice, implied volatility at the market open reflects forward-looking expectations and may diverge materially from realized historical volatility, particularly around scheduled macro events. To the extent that implied volatility is systematically lower under DDS conditions relative to aligned conditions — consistent with the mechanism proposed in Section 5 — the use of historical volatility may understate this difference and therefore introduce a conservative bias in the reported spread between conditions. Future work using real-time options data would permit a direct test of this hypothesis.

4. Results

4.1 Main Finding: Directional Disagreement vs. Alignment

Table 1 presents the primary results comparing option returns under directional disagreement (DDS = 1) and aligned (DDS = 0) conditions. The disagreement condition produces an average return of +32.03% per trade with a win rate of 82.3% and a Sharpe ratio of 15.04, compared to +7.59%, 50.7%, and 3.50 respectively under the aligned condition. The difference in mean returns (+24.44 percentage points) is statistically significant at conventional levels ($t = 49.67$, $p < 0.0001$).

While the magnitude of returns is economically large, these values should not be interpreted as directly achievable in live trading without further validation.

Table 1: Main Results — DDS Condition vs. Aligned Condition

Condition	N	Win Rate	Avg Return	Sharpe Ratio
DDS = 1 (Disagreement)	8,317	82.3%	+32.03%	15.04
DDS = 0 (Aligned)	11,284	50.7%	+7.59%	3.50
Difference	—	+31.6pp	+24.44pp	—

Note: t-statistic = 49.67, $p < 0.0001$. 95% CI for mean difference: [+23.5%, +25.4%]. Permutation p-value: < 0.001 (0 of 1,000 permutations exceeded observed difference). All returns net of 7% spread and 2% slippage.

4.2 Entry Time Decay

Table 2 reports average option returns and Sharpe ratios for four entry windows, ranging from the 9:31 AM market open to 10:30 AM. Across all 19,601 simulated trades (pooling both DDS conditions), returns decline monotonically with entry delay. This pattern is consistent with the hypothesis that option mispricing at the open is transient: as the market incorporates additional information through the opening minutes, the informational advantage embedded in the DDS signal dissipates.

Table 2: Returns by Entry Time (All Trades, Pooled)

Entry Time	N	Win Rate	Avg Return	Sharpe Ratio
9:31 AM (Open)	19,601	64%	+17.96%	7.87
9:46 AM	19,660	59%	+12.44%	5.87
10:00 AM	19,672	56%	+9.80%	4.84

10:30 AM	19,704	52%	+5.91%	3.23
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Note: OPEN_931 is the locked production entry for the live forward-testing period. Monotonic decay across all entry windows is consistent across VIX buckets and regimes.

4.3 Out-of-Sample Validation

To assess whether the DDS effect is a statistical artifact of the full sample, we split the dataset into an in-sample period (2020–2023) and an out-of-sample period (2024–2026). The out-of-sample period was not used in signal construction or parameter selection. Table 3 presents results for each period.

The DDS effect persists in the out-of-sample period with only modest degradation: win rate declines by 3.1 percentage points and mean return declines by 0.56 percentage points. This level of out-of-sample stability is consistent with a genuine structural effect rather than in-sample overfitting.

Table 3: In-Sample vs. Out-of-Sample Performance

Period	Condition	Win Rate	Avg Return
In-Sample (2020–2023)	DDS = 1	83.9%	+32.32%
In-Sample (2020–2023)	DDS = 0	51.2%	+7.91%
Out-of-Sample (2024–2026)	DDS = 1	80.8%	+31.76%
Out-of-Sample (2024–2026)	DDS = 0	50.1%	+7.21%

Note: Out-of-sample period not used in any signal construction or parameter selection. OOS degradation: -3.1pp win rate, -0.56pp mean return.

5. Proposed Mechanism

We propose two complementary explanations for the DDS effect. The first is rooted in options pricing dynamics. When a stock closes higher on day $t-1$ but opens flat or lower on day t , the implied volatility embedded in at-the-money call options at the open is likely to be suppressed relative to the aligned condition. Reduced opening-gap magnitude signals lower realized intraday volatility expectations, resulting in relatively cheaper call premiums. If the prior day's momentum is directionally persistent — as the literature on intermediate-horizon momentum suggests — then purchasing cheaper calls on an underlying with sustained upward bias produces favorable risk-adjusted returns.

The second explanation draws on market microstructure. Prior-day winners that gap down or open flat at the subsequent open may represent episodes in which overnight supply from profit-taking temporarily outweighs demand. As intraday trading progresses, latent buying interest from momentum-oriented participants and short-covering activity reasserts itself, driving the price back toward or beyond the prior close. ATM call options purchased at the 9:31 AM open capture the full extent of this intraday recovery at leveraged magnitude.

An alternative interpretation is that the DDS signal captures a short-term dislocation between realized price movement and implied expectations embedded in option pricing, resulting in temporary mispricing conditions favorable to directional option buyers. Distinguishing between these mechanisms empirically

requires real-time implied volatility data and is left for future research.

6. Robustness Checks

6.1 Permutation Test and Multiple Testing

To assess whether the observed return differential between $DDS = 1$ and $DDS = 0$ conditions could arise by chance, we conduct a permutation test in which the DDS label is randomly reassigned across all trades 1,000 times. Under the null hypothesis of no systematic difference, the permuted mean return differential should be distributed symmetrically around zero.

The observed difference of +24.44 percentage points falls entirely outside the permutation null distribution, which ranges from -1.46% to +1.63%. The permutation p-value is < 0.001 (zero of 1,000 permutations exceeded the observed value), providing strong evidence against the null hypothesis of no effect.

With respect to multiple testing, we note that the DDS signal emerged as the dominant feature from a broader feature importance analysis prior to statistical testing. The XGBoost model trained on a candidate set of 112 features assigned Ret1d a gain-based importance of 41.8% and OpeningGap an importance of 39.3%, together accounting for 81.1% of total predictive power. The remaining 110 features combined for 18.9%. Given that the signal was identified through a data-driven importance ranking rather than exhaustive hypothesis testing across all 112 variables, the multiple testing concern is partially mitigated. Nonetheless, independent replication on a separate dataset would provide stronger confirmation.

6.2 Subperiod Stability

Table 4 presents $DDS = 1$ versus $DDS = 0$ return differentials disaggregated across four distinct market regimes within the sample period. The effect is statistically significant in all three regimes for which sample size is adequate. The COVID-19 period (March–April 2020) yields too few $DDS = 1$ observations ($n = 14$) to support inference and is excluded from the statistical comparison.

Table 4: DDS Effect by Market Subperiod

Subperiod	DDS=1 Avg	DDS=0 Avg	Difference	p-value
COVID-19 (Mar–Apr 2020)	+28.1%	—	—	0.43 ($n=14$, inconclusive)
Rate Hike Selloff (2022)	+34.1%	+6.6%	+27.5pp	< 0.0001
Bull Market (2023)	+31.7%	+6.8%	+24.9pp	< 0.0001
Tariff Selloff (2025–26)	+33.8%	+9.2%	+24.6pp	< 0.0001

Note: p-values from two-sample t-tests. COVID-19 subperiod excluded from statistical comparison due to insufficient sample size in the $DDS = 1$ cell.

6.3 Feature Importance Validation

Table 5 presents the top-ranked features from the XGBoost importance analysis. The concentration of predictive power in two variables — prior-day return and opening gap — is notable given the breadth of

the candidate feature set (112 variables spanning momentum, volatility, regime, calendar, and sector signals). The logistic regression model trained on the same features achieves 86.3% accuracy versus 86.9% for XGBoost, a difference of 0.6 percentage points. Given this negligible performance gap, the simpler logistic regression specification is preferred for production use in accordance with the principle of parsimony.

Table 5: XGBoost Feature Importance (Top Features)

Feature	XGBoost Gain Importance	Interpretation
Ret1d	41.8%	Prior-day equity return
OpeningGap	39.3%	Same-day opening gap vs. prior close
All Others	18.9%	112 remaining candidate features combined

Note: Importance metric is gain (average improvement in model accuracy per split). XGBoost accuracy = 86.9%; Logistic Regression accuracy = 86.3%.

7. Conclusion

This paper documents a consistent and statistically significant pattern in simulated short-term equity option returns: at-the-money calls purchased at the market open under directional disagreement conditions — where the prior day's return is positive but the same-day opening gap is flat or negative — produce substantially higher average returns and win rates than calls purchased under aligned conditions.

The effect is robust to out-of-sample testing, permutation analysis, and disaggregation across four distinct market regimes. Feature importance analysis indicates that just two variables — prior-day return and opening gap — account for 81% of all predictive power across a 112-variable candidate set, suggesting that the interaction is a first-order predictor rather than an artifact of model complexity.

The proposed mechanism — cheaper options pricing at the open when a prior winner fails to extend its gains — is mechanistically consistent with both the options pricing literature on volatility mispricing and the microstructure literature on short-horizon reversal dynamics. These interpretations remain to be formally tested with real-time implied volatility data.

A key open question is whether the observed statistical edge survives realistic execution constraints, including adverse selection, fill quality, and dynamic spread behavior at the market open. Ongoing forward testing, which began in April 2026, is intended to address these considerations. Future versions of this paper will incorporate live execution results and, if sufficient data are available, an expanded analysis using real-time implied volatility as a direct input to the signal construction.

Disclaimer: This paper is intended for research purposes only and does not constitute investment advice or a solicitation to trade securities or derivatives. All results are based on simulated execution and should not be interpreted as indicative of future trading performance.

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